Early-Pandemic COVID-19 Epidemiologic Analysis

Will Maberry

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```
library(ggplot2)
```

1 Study Data

This study uses a 10,000-record sample of reported COVID-19 cases restricted to the early pre-vaccine study period from March 15 through December 10, 2020.

Data were downloaded from the CDC COVID-19 Case Surveillance Public Use Data through the CDC Socrata API.

A reproducible computational subsample was constructed by randomly selecting 20 calendar dates within the study period using a fixed random seed (42). For each selected date, up to 500 records were retrieved from a contiguous block of surveillance records, with the starting position selected reproducibly using a date-specific random seed. Records were ordered by their internal Socrata identifier before retrieval.

The resulting dataset contains 10,000 surveillance records and 12 variables. Since sampling was performed by date and contiguous record blocks rather than by simple random sampling across all eligible cases, the sample should not be considered representative of all reported COVID-19 cases during the study period.

```
# Load COVID-19 surveillance sample
covid <- read.csv(
    "Data/covid_early_pandemic_sample_10k.csv",
    stringsAsFactors = FALSE
)

dim(covid)
```

```
[1] 10000    12
```

2 Analytic Cohort

The analysis is restricted to cases with known exposure and outcome status. Exposure is defined by `medcond_yn`, indicating whether a pre-existing medical condition was reported. The outcome is defined by `hosp_yn`, indicating a hospitalization was reported.

`medcond_yn` and `hosp_yn` are coded as {`Yes`, `No`, `Unknown`, `Missing`}. Only `Yes` and `No` records for both variables were retained in the exposure-outcome complete-case cohort.

```r
analysis <- covid[
    covid$medcond_yn %in% c("Yes", "No") &
    covid$hosp_yn %in% c("Yes", "No"),
]

dim(analysis)
```

```
[1] 1464   12
```

The resulting exposure-outcome complete-case cohort contained 1,464 cases (14.6% of the original sample). The remaining 8,536 cases (85.4%) were excluded because exposure status, hospitalization status, or both were not recorded as Yes or No.

```r
# Examine exposure and outcome reporting completeness
table(covid$medcond_yn, useNA = "ifany")
```

```
Missing      No Unknown     Yes
   7358     973     826     843
```

```r
table(covid$hosp_yn, useNA = "ifany")
```

```
Missing      No Unknown     Yes
   3037    4720    1220    1023
```

```r
# Examine joint reporting patterns
table(
    covid$medcond_yn,
    covid$hosp_yn,
    useNA = "ifany"
)
```

```
          Missing   No Unknown  Yes
  Missing    2838 3397     470  653
  No          109  728     109   27
  Unknown      43  136     554   93
  Yes          47  459      87  250
```

```r
# Percentage with known medical-condition status
known_medical_condition = mean(covid$medcond_yn %in% c("Yes", "No")) * 100

# Percentage with known hospitalization status
known_hospitalization = mean(covid$hosp_yn %in% c("Yes", "No")) * 100

# Percentage with BOTH exposure and outcome known
known_both = mean(
    covid$medcond_yn %in% c("Yes", "No") &
    covid$hosp_yn %in% c("Yes", "No")
) * 100

# Display reporting completeness
round(c(
    Known_Hospitalization = known_hospitalization,
    Known_Medical_Condition = known_medical_condition,
    Known_Medical_Condition_and_Hospitalization = known_both
), 3)
```

```
                      Known_Hospitalization
                                      57.43
                    Known_Medical_Condition
                                      18.16
Known_Medical_Condition_and_Hospitalization
                                      14.64
```

Reporting completeness differed substantially between variables. Hospitalization status was known for 57.43% of sampled records, while pre-existing medical condition status was known for only 18.16%. Both variables were known for 14.64% of sampled records.

Restricting the analysis to cases with known exposure and outcome status may introduce selection bias if the excluded cases differ systematically from the included cases. The potential for selection bias is a limitation of this analysis.

3 Crude Association

3.1 Exposure-Outcome Table

We first estimate the crude association between the exposure (reported pre-existing medical condition) and outcome (hospitalization).

```r
# Construct 2x2 table with exposed/hospitalized first
exposure_outcome <- table(
    Exposure = factor(
        analysis$medcond_yn,
        levels = c("Yes", "No")
    ),
    Hospitalized = factor(
        analysis$hosp_yn,
        levels = c("Yes", "No")
    )
)

# Display table with row and column totals
addmargins(exposure_outcome)
```

```
        Hospitalized
Exposure  Yes   No  Sum
     Yes  250  459  709
     No    27  728  755
     Sum  277 1187 1464
```

3.2 Risk Ratio and Risk Difference

Here a and b represent hospitalized and non-hospitalized exposed cases, respectively. Similarly, c and d represent hospitalized and non-hospitalized unexposed cases, respectively. The crude risk ratio (RR) and risk difference (RD) are defined as:

$$RR = \frac{a/(a+b)}{c/(c+d)} \qquad RD = \frac{a}{a+b} - \frac{c}{c+d}$$

Hospitalization risk is estimated separately among exposed and unexposed cases.

```r
# Extract cells from exposure-outcome table
a <- exposure_outcome["Yes", "Yes"]
b <- exposure_outcome["Yes", "No"]
c <- exposure_outcome["No", "Yes"]
d <- exposure_outcome["No", "No"]

# Hospitalization risks
risk_exposed <- a / (a + b)
risk_unexposed <- c / (c + d)
```

```r
# Effect measures
risk_ratio <- risk_exposed / risk_unexposed
risk_difference <- risk_exposed - risk_unexposed

# Display hospitalization risks and effect measures
round(c(
    Risk_Exposed = risk_exposed,
    Risk_Unexposed = risk_unexposed,
    Risk_Ratio = risk_ratio,
    Risk_Difference = risk_difference
), 3)
```

```
   Risk_Exposed  Risk_Unexposed      Risk_Ratio Risk_Difference
          0.353           0.036           9.860           0.317
```

Among cases in the complete-case analytic cohort, hospitalization was reported for 35.3% of individuals with a reported pre-existing medical condition and 3.6% of individuals without one. The crude risk ratio was 9.86, indicating that the proportion hospitalized was approximately 9.86 times as high among individuals with reported medical conditions. The crude risk difference was 31.7 percentage points, representing the absolute difference in hospitalization proportions between the two groups.

3.3 95% Confidence Interval for Risk Ratio

A 95% confidence interval (CI) for the crude risk ratio was calculated using hte normal approximation to the log-tranformed risk ratio. The standard error of the log risk ratio is estimated as:

$$SE[\ln(RR)] = \sqrt{\frac{1}{a} - \frac{1}{a+b} + \frac{1}{c} - \frac{1}{c+d}}$$

The confidence interval is calculated on the logarithmic scale and then exponentiated to return to the risk-ratio scale.

$$95\%CI_{RR} = \exp\left[\ln(RR) \pm 1.96 \times SE[\ln(RR)]\right]$$

```r
# Standard error of the log risk ratio
se_log_rr <- sqrt(
    (1 / a) - (1 / (a + b)) +
    (1 / c) - (1 / (c + d))
)

# Calculate confidence limits on log scale
log_rr <- log(risk_ratio)

log_rr_lower <- log_rr - (1.96 * se_log_rr)
log_rr_upper <- log_rr + (1.96 * se_log_rr)

# Transform confidence limits back to risk-ratio scale
rr_lower <- exp(log_rr_lower)
rr_upper <- exp(log_rr_upper)

round(c(
    RR = risk_ratio,
    Lower_95_CI = rr_lower,
    Upper_95_CI = rr_upper
), 3)
```

```
         RR Lower_95_CI Upper_95_CI
      9.860       6.719      14.470
```

The crude risk ratio was 9.86 (95% CI: 6.72–14.47). The confidence interval excludes the null value of 1, indicating evidence of a positive crude association between reported pre-existing medical condition and hospitalization.

3.4 95% Confidence Interval for Risk Difference

A 95% confidence interval for the crude risk difference was calculated using the normal approximation for the difference between two independent proportions. The standard error is estimated as:

$$SE(RD) = \sqrt{\frac{R_1(1-R_1)}{n_1} + \frac{R_0(1-R_0)}{n_0}}$$

Here, R_1 and R_0 represent the observed hospitalization proportions among exposed and unexposed cases, respectively, while n_1 and n_0 represent the corresponding group sizes.

The confidence interval is calculated directly on the risk-difference scale:

$$95\%CI_{RD} = RD \pm 1.96 \times SE(RD)$$

```
# Group sample sizes
n_exposed <- a + b
n_unexposed <- c + d

# Standard error of risk difference
se_rd <- sqrt(
    (risk_exposed * (1 - risk_exposed) / n_exposed) +
    (risk_unexposed * (1 - risk_unexposed) / n_unexposed)
)

# Calculate confidence limits
rd_lower <- risk_difference - (1.96 * se_rd)
rd_upper <- risk_difference + (1.96 * se_rd)

round(c(
    RD = risk_difference,
    Lower_95_CI = rd_lower,
    Upper_95_CI = rd_upper
), 3)
```

```
        RD Lower_95_CI Upper_95_CI
     0.317       0.279       0.354
```

The crude risk difference was 31.7 percentage points (95% CI: 27.9–35.4 percentage points). The confidence interval excludes the null value of 0, indicating evidence of a positive crude difference in hospitalization proportions between reported pre-existing medical condition and hospitalization.

These estimates describe crude, unadjusted associations and do not account for potential confounding by age or sex. Both confidence intervals were calculated using the normal approximation, which may be inaccurate for small sample sizes or extreme proportions.

4 Descriptive Analysis

Age and sex were hypothesized to be potential confounders based on the directed acyclic graph (DAG) and prior literature review. Before fitting adjusted models, their distributions were

examined across exposure groups, along with the observed proportion hospitalized within each age and sex category.

4.1 Age

4.1.1 Age Distribution by Exposure

The age distribution was examined separately among cases with and without reported pre-existing medical conditions.

```
# Age distribution by exposure status
age_exposure <- table(
    analysis$age_group,
    analysis$medcond_yn
)

# Counts
age_exposure
```

```
                No Yes
  0 - 9 Years   134  15
  10 - 19 Years  65  23
  20 - 29 Years 171  78
  30 - 39 Years 164  98
  40 - 49 Years  52  46
  50 - 59 Years 104  92
  60 - 69 Years  51 224
  70 - 79 Years   9  49
  80+ Years       5  84
```

```
# Column percentages: age distribution within each exposure group
round(
    prop.table(age_exposure, margin = 2) * 100,
    1
)
```

```
                 No  Yes
  0 - 9 Years   17.7  2.1
  10 - 19 Years  8.6  3.2
```

```
  20 - 29 Years 22.6 11.0
  30 - 39 Years 21.7 13.8
  40 - 49 Years  6.9  6.5
  50 - 59 Years 13.8 13.0
  60 - 69 Years  6.8 31.6
  70 - 79 Years  1.2  6.9
  80+ Years      0.7 11.8
```

Column percentages represent the age distribution within each exposure group, $P(Age \mid E)$. Each exposure column sums to 100%.

```
# Define age groups representing individuals aged 60+
older_groups <- c(
    "60 - 69 Years",
    "70 - 79 Years",
    "80+ Years"
)

# Calculate percentage aged 60+ within each exposure group
older_percent <- colSums(
    age_exposure[older_groups, , drop = FALSE]
) / colSums(age_exposure) * 100

round(older_percent, 1)
```

```
  No  Yes
 8.6 50.4
```

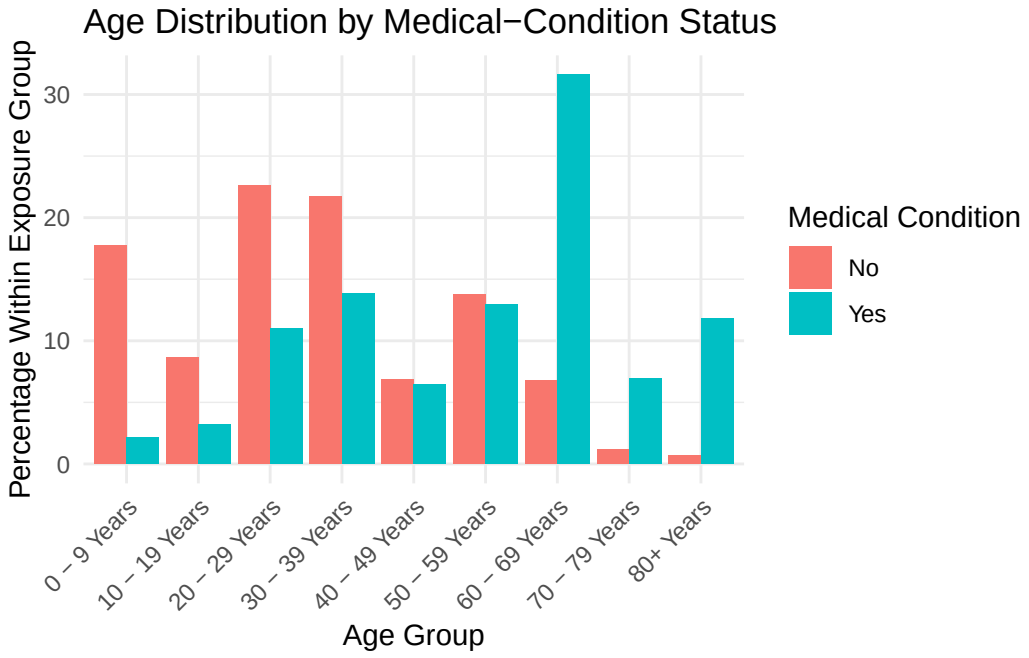
The age distribution differed substantially between exposure groups. Among cases with reported pre-existing medical conditions, 50.4% were aged 60 years or older, compared with 8.6% among cases without reported conditions. Older individuals were disproportionately represented in the exposed group, suggesting that age may contribute to the observed crude association between reported pre-existing medical condition and hospitalization.

```
# Convert column percentages to a data frame for plotting
age_exposure_df <- as.data.frame(
    prop.table(age_exposure, margin = 2) * 100
)

names(age_exposure_df) <- c(
    "Age",
```

```
    "Medical_Condition",
    "Percentage"
)

# Plot age distribution within each exposure group
ggplot(
    age_exposure_df,
    aes(
        x = Age,
        y = Percentage,
        fill = Medical_Condition
    )
) +
    geom_col(position = "dodge") +
    labs(
        title = "Age Distribution by Medical-Condition Status",
        x = "Age Group",
        y = "Percentage Within Exposure Group",
        fill = "Medical Condition"
    ) +
    theme_minimal() +
    theme(
        axis.text.x = element_text(angle = 45, hjust = 1)
    )
```



4.1.2 Hospitalization by Age

The proportion hospitalized was calculated within each age group to examine whether observed hospitalization proportion varies by age.

```
# Hospitalization status by age group
age_outcome <- table(
    analysis$age_group,
    analysis$hosp_yn
)

# Counts
age_outcome
```

```
                No Yes
  0 - 9 Years   145   4
  10 - 19 Years  86   2
  20 - 29 Years 240   9
  30 - 39 Years 238  24
  40 - 49 Years  87  11
  50 - 59 Years 184  12
```

```
  60 - 69 Years 127 148
  70 - 79 Years  27  31
  80+ Years      53  36
```

```r
# Row percentages: hospitalization status within each age group
round(
    prop.table(age_outcome, margin = 1) * 100,
    1
)
```

```
                 No  Yes
  0 - 9 Years   97.3  2.7
  10 - 19 Years 97.7  2.3
  20 - 29 Years 96.4  3.6
  30 - 39 Years 90.8  9.2
  40 - 49 Years 88.8 11.2
  50 - 59 Years 93.9  6.1
  60 - 69 Years 46.2 53.8
  70 - 79 Years 46.6 53.4
  80+ Years     59.6 40.4
```

Row percentages represent hospitalization status within each age group, $P(Y \mid Age)$. Each age-group row sums to 100%.

```r
# Calculate percentage hospitalized within each age group
age_hosp_percent <- prop.table(
    age_outcome,
    margin = 1
)[, "Yes"] * 100

round(age_hosp_percent, 1)
```

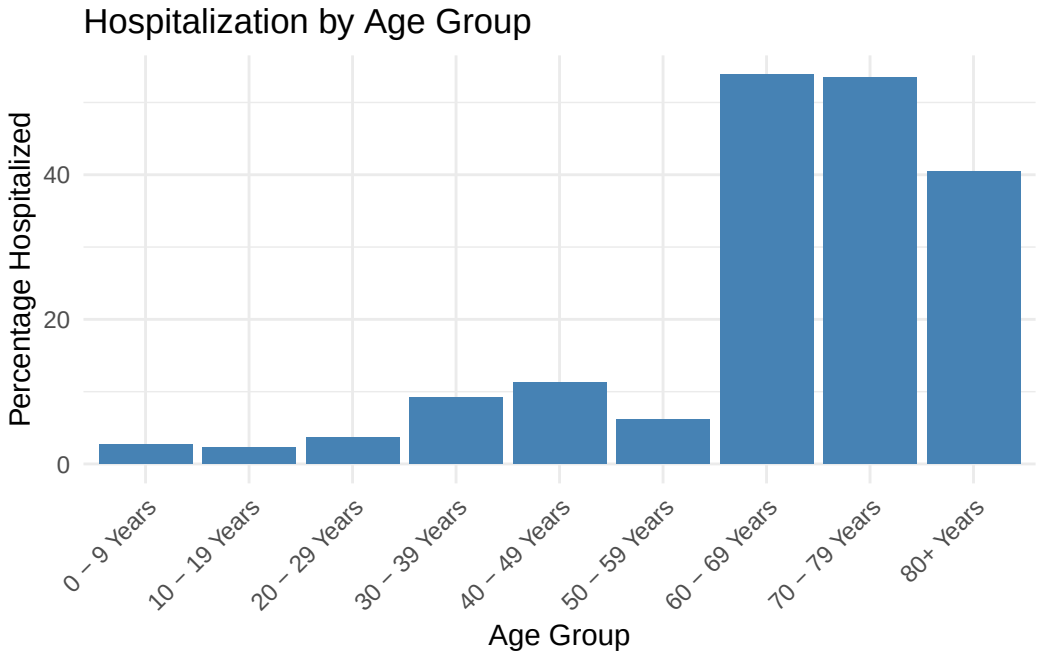
```
  0 - 9 Years 10 - 19 Years 20 - 29 Years 30 - 39 Years 40 - 49 Years
          2.7           2.3           3.6           9.2          11.2
50 - 59 Years 60 - 69 Years 70 - 79 Years     80+ Years
          6.1          53.8          53.4          40.4
```

Hospitalization proportions varied substantially across age groups. Among cases aged 60–69 years, 53.8% were hospitalized, compared with 3.6% among cases aged 20–29 years. The proportion hospitalized among cases aged 80 years or older was 40.4%, indicating that the

observed pattern did not increase consistently across all age categories. These findings suggest a strong descriptive association between age and hospitalization within the analytic cohort.

```
# Convert hospitalization percentages into a data frame for plotting
age_hosp_df <- data.frame(
    Age = names(age_hosp_percent),
    Percentage = as.numeric(age_hosp_percent)
)

# Plot hospitalization percentage within each age group
ggplot(
    age_hosp_df,
    aes(
        x = Age,
        y = Percentage
    )
) +
    geom_col(fill = "steelblue") +
    labs(
        title = "Hospitalization by Age Group",
        x = "Age Group",
        y = "Percentage Hospitalized"
    ) +
    theme_minimal() +
    theme(
        axis.text.x = element_text(angle = 45, hjust = 1)
    )
```



4.2 Sex

4.2.1 Sex by Exposure

The sex distribution was examined separately among cases with and without reported pre-existing medical conditions.

```
# Sex distribution by exposure status
sex_exposure <- table(
    analysis$sex,
    analysis$medcond_yn
)

# Counts
sex_exposure
```

```
           No Yes
  Female  368 327
  Male    386 381
  Missing   1   0
  Unknown   0   1
```

```r
# Column percentages: sex distribution within each exposure group
round(
    prop.table(sex_exposure, margin = 2) * 100,
    1
)
```

```
           No  Yes
  Female  48.7 46.1
  Male    51.1 53.7
  Missing  0.1  0.0
  Unknown  0.0  0.1
```

Column percentages represent the sex distribution within each exposure group, $P(Sex \mid E)$. Each exposure column sums to 100%.

```r
# Calculate percentage of males within each exposure group
male_percent <- prop.table(
    sex_exposure,
    margin = 2
)["Male", ] * 100

round(male_percent, 1)
```

```
  No  Yes
51.1 53.7
```

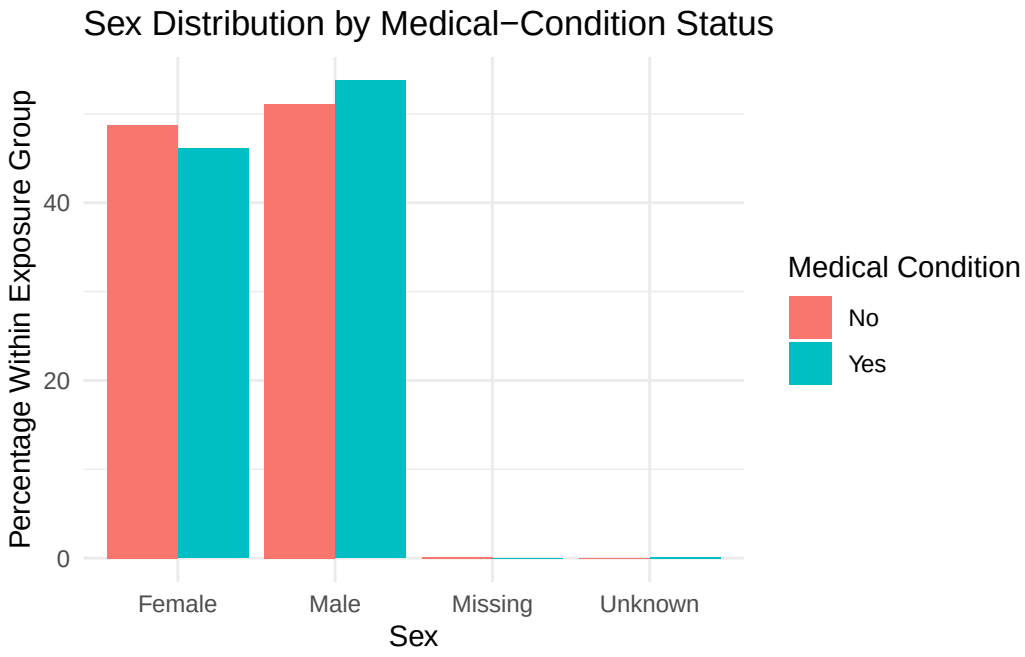
The sex distribution was similar between exposure groups. Males accounted for 53.7% of cases with reported pre-existing medical conditions, compared with 51.1% of cases without reported conditions. These relatively small differences suggest that sex may contribute less to the crude exposure-outcome association than age, although this descriptive comparison alone does not establish the extent of confounding.

```r
# Convert column percentages to a data frame for plotting
sex_exposure_df <- as.data.frame(
    prop.table(sex_exposure, margin = 2) * 100
)

names(sex_exposure_df) <- c(
    "Sex",
```

```
    "Medical_Condition",
    "Percentage"
)

# Plot sex distribution within each exposure group
ggplot(
    sex_exposure_df,
    aes(
        x = Sex,
        y = Percentage,
        fill = Medical_Condition
    )
) +
    geom_col(position = "dodge") +
    labs(
        title = "Sex Distribution by Medical-Condition Status",
        x = "Sex",
        y = "Percentage Within Exposure Group",
        fill = "Medical Condition"
    ) +
    theme_minimal()
```



4.2.2 Hospitalization by Sex

The proportion hospitalized was calculated within each sex category to examine differences in hospitalization risk between males and females.

```
# Hospitalization status by sex
sex_outcome <- table(
    analysis$sex,
    analysis$hosp_yn
)

# Counts
sex_outcome
```

```
           No Yes
  Female  573 122
  Male    612 155
  Missing   1   0
  Unknown   1   0
```

```
# Row percentages: hospitalization status within each sex group
round(
    prop.table(sex_outcome, margin = 1) * 100,
    1
)
```

```
             No   Yes
  Female   82.4  17.6
  Male     79.8  20.2
  Missing 100.0   0.0
  Unknown 100.0   0.0
```

Row percentages represent hospitalization status within each sex category, $P(Y \mid Sex)$. Each sex row sums to 100%.

```
# Calculate percentage hospitalized within each sex category
sex_hosp_percent <- prop.table(
    sex_outcome,
    margin = 1
```

```
)[, "Yes"] * 100

round(sex_hosp_percent, 1)
```

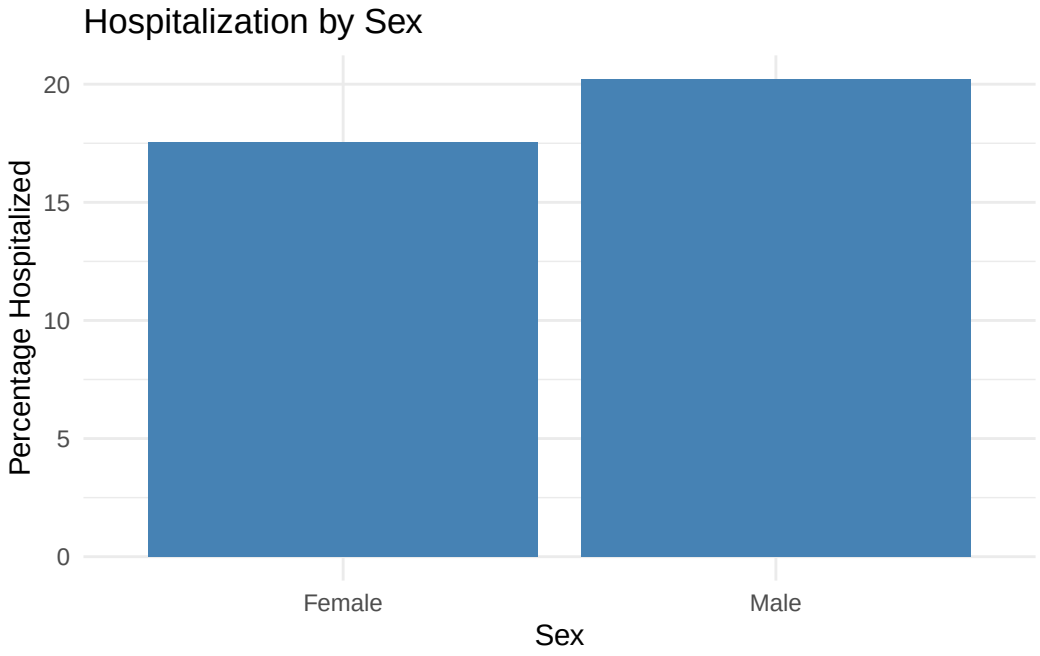
```
 Female    Male Missing Unknown
   17.6    20.2     0.0     0.0
```

Hospitalization proportions differed modestly between males and females. Among male cases, 20.2% were hospitalized, compared with 17.6% among female cases. These results suggest a relatively small descriptive association between sex and hospitalization within the analytic cohort.

```
# Convert hospitalization percentages into a data frame for plotting
sex_hosp_df <- data.frame(
    Sex = names(sex_hosp_percent),
    Percentage = as.numeric(sex_hosp_percent)
)

# Keep Female and Male categories for visualization
sex_hosp_df <- subset(
    sex_hosp_df,
    Sex %in% c("Female", "Male")
)

# Plot hospitalization percentage within each sex category
ggplot(
    sex_hosp_df,
    aes(
        x = Sex,
        y = Percentage
    )
) +
    geom_col(fill = "steelblue") +
    labs(
        title = "Hospitalization by Sex",
        x = "Sex",
        y = "Percentage Hospitalized"
    ) +
    theme_minimal()
```



4.3 Summary of Potential Confounders

The descriptive comparisons were used to examine the relationships of age and sex with both reported pre-existing medical conditions and hospitalization. Differences in these distributions may contribute to the observed crude association.

Age and sex were retained as adjustment variables based on the hypothesized causal structure and prior literature review rather than solely on the descriptive results. The following adjusted analysis examines the exposure-outcome association while accounting for these variables.

5 Adjusted Analysis

Age and sex were identified as potential confounders based on the hypothesized causal structure and prior literature review. To examine the association between reported pre-existing medical conditions and hospitalization while accounting for these variables, a multivariable logistic regression model was fitted.

5.1 Adjusted Analytic Cohort

The adjusted analysis was restricted to cases with known exposure, outcome, age, and sex status. Two records in the exposure-outcome complete-case cohort had sex coded as `Missing` or `Unknown` and have been excluded.

```r
# Restrict analysis to cases with recorded Female or Male sex
adjusted_analysis <- analysis[
    analysis$sex %in% c("Female", "Male") &
    !is.na(analysis$age_group) &
    analysis$age_group != "Missing" &
    analysis$age_group != "Unknown",
]

# Display adjusted cohort size
nrow(adjusted_analysis)
```

```
[1] 1462
```

The resulting adjusted-analysis cohort contained 1462 cases.

The outcome, axposure, and adjustment variables were encoded as categorical factors to define their reference categories explicitly.

```r
# Hospitalization: No is the reference, Yes is the event
adjusted_analysis$hospitalized <- factor(
    adjusted_analysis$hosp_yn,
    levels = c("No", "Yes")
)

# Medical condition: No is the reference
adjusted_analysis$medcond <- factor(
    adjusted_analysis$medcond_yn,
    levels = c("No", "Yes")
)

# Sex: Female is the reference
adjusted_analysis$sex_factor <- factor(
    adjusted_analysis$sex,
    levels = c("Female", "Male")
)

# Age: 0-9 years is the reference
adjusted_analysis$age_factor <- factor(
    adjusted_analysis$age_group,
    levels = c(
        "0 - 9 Years",
        "10 - 19 Years",
```

```
        "20 - 29 Years",
        "30 - 39 Years",
        "40 - 49 Years",
        "50 - 59 Years",
        "60 - 69 Years",
        "70 - 79 Years",
        "80+ Years"
    )
)

# Verify reference categories
lapply(
    adjusted_analysis[
        c("hospitalized", "medcond", "age_factor", "sex_factor")
    ],
    levels
)
```

```
$hospitalized
[1] "No"  "Yes"

$medcond
[1] "No"  "Yes"

$age_factor
[1] "0 - 9 Years"   "10 - 19 Years" "20 - 29 Years" "30 - 39 Years"
[5] "40 - 49 Years" "50 - 59 Years" "60 - 69 Years" "70 - 79 Years"
[9] "80+ Years"

$sex_factor
[1] "Female" "Male"
```

```
# Verify that all model variables are complete
colSums(is.na(
    adjusted_analysis[
        c("hospitalized", "medcond", "age_factor", "sex_factor")
    ]
))
```

```
hospitalized      medcond   age_factor   sex_factor
           0            0            0            0
```

Hospitalization was modeled with `Yes` as the event and `No` as the reference category. Reported pre-existing medical-condition status used `No` as the reference category. Sex used `Female` as the reference category, while age used `0-9 Years` as the reference group.

5.2 Logistic Regression

A multivariable logistic regression model was fit with hospitalization as the binary outcome, reported pre-existing medical condition as the exposure, and age group and sex as adjustment variables.

```
# Fit logistic regression adjusting for age and sex
logistic_model <- glm(
    hospitalized ~ medcond + age_factor + sex_factor,
    data = adjusted_analysis,
    family = binomial(link = "logit")
)

# Examine model coefficients and standard errors
summary(logistic_model)
```

```
Call:
glm(formula = hospitalized ~ medcond + age_factor + sex_factor,
    family = binomial(link = "logit"), data = adjusted_analysis)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)             -4.180e+00  5.354e-01  -7.807 5.86e-15 ***
medcondYes               1.874e+00  2.319e-01   8.079 6.50e-16 ***
age_factor10 - 19 Years -5.267e-01  8.929e-01  -0.590 0.555311
age_factor20 - 29 Years -1.615e-01  6.273e-01  -0.258 0.796792
age_factor30 - 39 Years  6.146e-01  5.667e-01   1.084 0.278178
age_factor40 - 49 Years  7.327e-01  6.196e-01   1.183 0.237001
age_factor50 - 59 Years  2.942e-05  6.066e-01   0.000 0.999961
age_factor60 - 69 Years  2.624e+00  5.424e-01   4.838 1.31e-06 ***
age_factor70 - 79 Years  2.567e+00  5.977e-01   4.295 1.75e-05 ***
age_factor80+ Years      1.964e+00  5.850e-01   3.358 0.000785 ***
sex_factorMale           2.805e-01  1.824e-01   1.538 0.123996
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1419.45  on 1461  degrees of freedom
Residual deviance:  947.29  on 1451  degrees of freedom
AIC: 969.29

Number of Fisher Scoring iterations: 6
```

Logistic regression estimates associates on the log-odds scale. As a result, the exposure coefficient must be exponentiated to obtain an adjusted odds ratio.

```r
# Extract the exposure coefficient (log odds ratio)
exposure_beta <- coef(logistic_model)["medcondYes"]

# Convert the coefficient to an adjusted odds ratio
adjusted_or <- exp(exposure_beta)

# Display adjusted odds ratio
round(adjusted_or, 3)
```

```
medcondYes
     6.513
```

A 95% confidence interval for the adjusted odds ratio was calculated using the standard error of the exposure coefficient. Confidence limits were calculated on the log-odds scale and exponentiated to return to the odds-ratio scale.

$$\exp\left[\hat{\beta}_E \pm 1.96 \times SE(\hat{\beta}_E)\right]$$

```r
# Extract standard error of the exposure coefficient
exposure_se <- summary(logistic_model)$coefficients[
    "medcondYes",
    "Std. Error"
]

# Calculate 95% confidence limits on the log-odds scale
or_log_lower <- exposure_beta - 1.96 * exposure_se
or_log_upper <- exposure_beta + 1.96 * exposure_se

# Transform confidence limits to the odds-ratio scale
or_lower <- exp(or_log_lower)
```

```r
or_upper <- exp(or_log_upper)

# Display adjusted odds ratio and confidence interval
round(
    c(
        Adjusted_OR = adjusted_or,
        Lower_95_CI = or_lower,
        Upper_95_CI = or_upper
    ),
    3
)
```

```
Adjusted_OR.medcondYes Lower_95_CI.medcondYes Upper_95_CI.medcondYes
                 6.513                  4.134                 10.260
```

The fitted model estimated an adjusted odds ratio of 6.51 (95% CI: 4.13–10.26) for reported pre-existing medical conditions, controlling for age group and sex. This indicates that the estimated odds of hospitalization were higher among cases with reported pre-existing medical conditions than among cases without, after accounting for age and sex.

5.2.1 Interpretation of Logistic Regression

Logistic regression models the log odds of a binary outcome as a function of the exposure and adjustment varialbes. The fitted model can be written as:

$$\beta_0 + \beta_E E_i + \sum_{k=1}^{8} \beta_{A_k} A_{ik} + \beta_S S_i,$$

where

$$p_i = P(Y_i = 1 \mid E_i, A_i, S_i).$$

Here, E_i represents reported pre-existing medical condition status, A_{ik} represents the eight indicator variables from the non-reference age groups, and S_i represents sex.

The exposure coefficient β_E represents the difference in conditional log odds of hospitalization between cases with and without reported pre-existing medical conditions, holding age group and sex constant.

Exponentiating this coefficient gives the adjusted odds ratio:

$$OR_{\text{adjusted}} = e^{\beta_E},$$

This odds ratio compares hospitalization odds between exposure groups at the same values of age and sex. The model assumes that the exposure odds ratio is constant across age and sex categories

5.3 Adjusted Odds Ratio

A 95% Wald confidence interval was calculated on the log-odds scale using the estimated coefficient and its standard error, then exponentiated to obtain confidence limits on the odds-ratio scale.

$$OR_{\text{adjusted}} = e^{\hat{\beta}_E}$$

$$95\%CI_{OR} = \exp(\hat{\beta}_E \pm 1.96 \times SE(\hat{\beta}_E))$$

```
# Extract the exposure coefficient (log odds ratio)
beta <- coef(logistic_model)["medcondYes"]

# Exponentiate to obtain the adjusted odds ratio
adjusted_or <- exp(beta)

# Extract the coefficient variance and calculate its standard error
se <- sqrt(vcov(logistic_model)["medcondYes", "medcondYes"])

# Calculate 95% Wald confidence interval on the odds-ratio scale
or_ci <- exp(beta + c(-1, 1) * 1.96 * se)

# Display adjusted odds ratio and confidence interval
round(
    c(
        Adjusted_OR = adjusted_or,
        Lower_95_CI = or_ci[1],
        Upper_95_CI = or_ci[2]
    ),
    3
)
```

```
Adjusted_OR.medcondYes          Lower_95_CI          Upper_95_CI
                 6.513                4.134               10.260
```

The fitted model estimate an adjusted odds ratio of 6.51 (95% CI: 4.13–10.26) for reported pre-existing medical conditions, conditional on age group and sex. This indicates that the estimated odds of hospitalization were higher among cases with reported pre-existing medical conditions than among cases without reported conditions, after accounting for age and sex.

5.4 Regression Standardization

Regression standardization was used to estimate hospitalization proportions for hte two exposure conditions while accounting for the age and sex distribution of the adjusted-analysis cohort.

This approach uses the fitted logistic regression model to predict hospitalization probabilities under two hypothetical exposure scenarios: one in which every case is assigned reported pre-existing medical conditions and another in which every case is assigned no reported pre-existing medical conditions.

Age and sex remain unchanged in both scenarios. This means the predicted probabilities are averages over the same obeserved distribution of age and sex, standardizing for these potential confounders.

5.4.1 Constructing Exposure Scenarios

Two copies of the adjusted-analysis cohort were created. Pre-existing medical condition status was set to `Yes` for every case in one copy and `No` for every case in the other copy. Age and sex were left unchanged in both copies.

```
# Create two copies of the adjusted cohort
everyone_exposed <- adjusted_analysis
everyone_unexposed <- adjusted_analysis

# Assign reported medical conditions to every case
everyone_exposed$medcond <- factor(
    "Yes",
    levels = levels(adjusted_analysis$medcond)
)

# Assign no reported medical conditions to every case
everyone_unexposed$medcond <- factor(
    "No",
```

```
    levels = levels(adjusted_analysis$medcond)
)
```

These scenarios are used to generate model-based predictions. They do not represent actual observed data, but rather hypothetical situations in which every case is assigned the same exposure status while retaining their observed age and sex.

5.4.2 Predicting Standardized Hospitalization Proportions

The fitted logistic regression model was used to predict hospitalization probabilities for each case under both exposure scenarios.

Setting `type = "response"` returns the predicted probabilities rather than predicted log odds

```
# Predict hospitalization probabilities under each exposure scenario
pred_exposed <- predict(
    logistic_model,
    newdata = everyone_exposed,
    type = "response"
)

pred_unexposed <- predict(
    logistic_model,
    newdata = everyone_unexposed,
    type = "response"
)

# Average predicted probabilities across the adjusted cohort
adjusted_risk_exposed <- mean(pred_exposed)
adjusted_risk_unexposed <- mean(pred_unexposed)

# Display standardized hospitalization proportions
round(
    c(
        Exposed = adjusted_risk_exposed,
        Unexposed = adjusted_risk_unexposed
    ),
    4
)
```

```
  Exposed Unexposed
   0.2551    0.0669
```

The standardized hospitalization proportion under reported pre-existing medical conditions was 25.5%, compared with 6.7% under no reported medical conditions.

These are model-based estimates averaged across the observed age and sex distribution of the adjusted-analysis cohort, rather than the directly observed hospitalization proportions within each exposure group.

5.5 Adjusted Risk Ratio and Risk Difference

The standardized hospitalization proportions were used to calculate an adjusted risk ratio and risk difference.

The adjusted risk ratio compares the two standardized proportions:

$$RR_{\text{adjusted}} = \frac{\bar{p}E = 1}{\bar{p}E = 0}$$

The adjusted risk difference represents their absolute difference:

$$RD_{\text{adjusted}} = \bar{p}E = 1 - \bar{p}E = 0$$

Where $\bar{p}E = 1$ and $\bar{p}E = 0$ are the mean model-predicted hospitalizaiton probabilities under the exposed and unexposed scenarios, respectively.

```
# Calculate adjusted risk ratio and risk difference
adjusted_rr <- adjusted_risk_exposed / adjusted_risk_unexposed
adjusted_rd <- adjusted_risk_exposed - adjusted_risk_unexposed

# Display adjusted association measures
round(
    c(
        Adjusted_RR = adjusted_rr,
        Adjusted_RD = adjusted_rd,
        Adjusted_RD_Percentage_Points = adjusted_rd * 100
    ),
    3
)
```

```
                  Adjusted_RR                   Adjusted_RD
                        3.816                         0.188
Adjusted_RD_Percentage_Points
                       18.825
```

The standardized adjusted risk ratio was 3.82, indicating that the model-estimated hospitalization proportion under reported pre-existing medical conditions was approximately 3.82 times the proportion under no reported medical conditions, after standardizing to the same age and sex distribution.

The adjusted risk difference was 18.8 percentage points, representing the absolute difference between the two standardized hospitalization proportions.

These estimates describe model-adjusted associations within the analytic cohort. They do not establish a causal effect of pre-existing medical conditions on hospitalization. In particular, adjustment for age and sex does not eliminate potential residual confounding, selection bias associated with missing surveillance data, or limitations arising from the sampling procedure.